



# Predicting Pet Adoption Speed in a Malaysia Shelter Using Logistic Regression Models

Fergus O'Brien  
20183119

Department of Mathematics and Statistics  
University of Limerick

Supervisor  
Kevin Burke

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# Abstract

The goal of this paper is to build the best predictive model for the “Train” dataset containing a large amount of observations ( 15,000) including tabular (traditional) and multimodal (image, text, and video-based metadata) data. This dataset originates from the Petfinder Prediction competition. Type of data/variables- Character, integer, numeric. This study was done using logistic regression models and multimodal feature integration (the process of combining data from multiple modalities- text, image metadata, tabular data. Doing this should give a more complete view and improve model performance. This allows the model to pick up on patterns that are only visible when other modalities interact, e.g. A pet with positive sentiment in the description and labeled as ”dog” with a high-quality photo may be more likely to be adopted. The best model built was using just the tabular data, giving an AIC score of 1496. Including the multimodal data in the model, the AIC score reached was 15,506, showing the model was more accurate without the images/text. (why?)

# Chapter 1

## Introduction

### 1.1 Introduction

Pet shelters worldwide get overcrowded and many pets are euthanized due to this. Increasing adoption rates would help reduce these numbers. Thanks to shelters releasing their datasets, we have an increasing amount of content for structured and unstructured data regarding these animals to be studied. Utilising machine learning techniques in this field is of growing importance to help animal welfare, helping to improve the likelihood of adoption for these animals.

The dataset being used in this study is from the Petfinder.my Adoption Prediction competition. <https://www.kaggle.com/competitions/petfinder-adoption-prediction/overview>. This dataset includes nearly 15,000 observations and combines structured tabular data (Age/Breed), sentiment score and magnitude from text, and image metadata. This type of data is known as multimodal data.

Predictive modelling used to rely predominantly on tabular data but there is more and more multimodal data available to study and methods to build models that include valuable information hidden in text or visual features. By unifying the types of data, subtle patterns may emerge (emotional tone, flattering photography) that can influence adoption rates.

This paper aims to explore logistic regression modelling related to the binary adoption speed variable by incorporating tabular and multimodal features. This variable shows whether a pet is adopted within 30 days or more. Splitting the variable like this gives us two groups similar in size to test the data on and the timeframe is reasonable. A comparison is made between the performance of models trained using only tabular data and models trained with sentiment analysis and metadata included as well. The model performance is evaluated using AIC scores



and classification accuracy + more? (ROC/AUC etc). So far, we have found that the tabular data alone outperforms the model including multimodal features. This means not all modalities necessarily enhance predictive accuracy.

## Chapter 2

# Literature review

Due to issues related to shelter overcrowding and euthanization research has been conducted to help identify factors that affect adoption rates.

The issues related to pet adoption, including shelter overcrowding and euthanasia, have prompted research initiatives aimed at identifying the factors that affect adoption rates and creating predictive models. This section examines research on the predictors of pet adoption, the use of tabular and multimodal data modelling, and the impact of advanced machine learning techniques such as XGBoost, deep learning, and large language models (LLMs) in enhancing adoption results.

### 2.1 Pet Adoption Predictors and Shelter Dynamics

It is important to understand the factors that relate to adoption rates while addressing overcrowded shelters and attempting to reduce euthanasia rates in these shelters. Reasons for overcrowding these shelters include giving up pets for adoption due to housing-related issues, personal circumstances like financial hardships and aggressive pet behaviour (Fantuzzi et al., 2010).

Certain characteristics that impact the likelihood of adoption include the pet's age, their breed, sterilization status and coat colour (Fantuzzi et al., 2010). An example of this is that purebred dogs are more likely to be adopted because the adopter may want a sense of predictability as regards their dog's size or energy levels (American Kennel Club, 2024). Another example of this is cats who approach people and demonstrate human-friendly behaviours tend to be adopted quicker than others (Tuozi, et al., 2021).

Data-driven approaches are used to determine these predictors. Predictive models such as

machine learning methods like CatBoost helped uncover qualities such as age, and brachycephaly i.e. short-snouted breeds. These tend to be key factors in determining adoption speed (Fantuzzi et al., 2010). In spite of this research, shelters continue to be overcrowded which indicates a gap between results and finding solutions.

## Chapter 3

# Exploratory Data Analysis

Below is a description of the data found in the Petfinder Prediction dataset.

Table 3.1: Descriptive statistics for the training dataset

| Variable      | Vars | N     | Mean     | SD      | Median | Trimmed  | MAD     | Min   | Max   | Range | Skew  |
|---------------|------|-------|----------|---------|--------|----------|---------|-------|-------|-------|-------|
| Type          | 1    | 14993 | 1.46     | 0.50    | 1      | 1.45     | 0.00    | 1     | 2     | 1     | 0.17  |
| Name*         | 2    | 14993 | 4211.58  | 2767.20 | 4278   | 4195.43  | 3642.75 | 1     | 9061  | 9060  | 0.01  |
| Age           | 3    | 14993 | 10.45    | 18.16   | 3      | 5.98     | 2.97    | 0     | 255   | 255   | 3.76  |
| Breed1        | 4    | 14993 | 265.27   | 60.06   | 266    | 279.51   | 60.79   | 0     | 307   | 307   | -2.22 |
| Breed2        | 5    | 14993 | 74.01    | 123.01  | 0      | 54.14    | 0.00    | 0     | 307   | 307   | 1.14  |
| Gender        | 6    | 14993 | 1.78     | 0.68    | 2      | 1.72     | 1.48    | 1     | 3     | 2     | 0.31  |
| Color1        | 7    | 14993 | 2.23     | 1.75    | 2      | 1.87     | 1.48    | 1     | 7     | 6     | 1.47  |
| Color2        | 8    | 14993 | 3.22     | 2.74    | 2      | 3.15     | 2.97    | 0     | 7     | 7     | 0.19  |
| Color3        | 9    | 14993 | 1.88     | 2.98    | 0      | 1.48     | 0.00    | 0     | 7     | 7     | 1.01  |
| MaturitySize  | 10   | 14993 | 1.86     | 0.55    | 2      | 1.84     | 0.00    | 1     | 4     | 3     | 0.01  |
| FurLength     | 11   | 14993 | 1.47     | 0.60    | 1      | 1.39     | 0.00    | 1     | 3     | 2     | 0.89  |
| Vaccinated    | 12   | 14993 | 1.73     | 0.67    | 2      | 1.66     | 1.48    | 1     | 3     | 2     | 0.37  |
| Dewormed      | 13   | 14993 | 1.56     | 0.70    | 1      | 1.45     | 0.00    | 1     | 3     | 2     | 0.85  |
| Sterilized    | 14   | 14993 | 1.91     | 0.57    | 2      | 1.89     | 0.00    | 1     | 3     | 2     | -0.01 |
| Health        | 15   | 14993 | 1.04     | 0.20    | 1      | 1.00     | 0.00    | 1     | 3     | 2     | 5.77  |
| Quantity      | 16   | 14993 | 1.58     | 1.47    | 1      | 1.20     | 0.00    | 1     | 20    | 19    | 4.60  |
| Fee           | 17   | 14993 | 21.26    | 78.41   | 0      | 2.15     | 0.00    | 0     | 3000  | 3000  | 8.92  |
| State         | 18   | 14993 | 41346.03 | 32.44   | 41326  | 41341.69 | 0.00    | 41324 | 41415 | 91    | 1.09  |
| RescuerID*    | 19   | 14993 | 2978.29  | 1619.09 | 3094   | 3004.17  | 2022.27 | 1     | 5595  | 5594  | -0.12 |
| VideoAmt      | 20   | 14993 | 0.06     | 0.35    | 0      | 0.00     | 0.00    | 0     | 8     | 8     | 9.46  |
| Description*  | 21   | 14993 | 6981.70  | 4028.88 | 6962   | 6973.40  | 5186.13 | 1     | 14033 | 14032 | 0.02  |
| PetID*        | 22   | 14993 | 7497.00  | 4328.25 | 7497   | 7497.00  | 5556.78 | 1     | 14993 | 14992 | 0.00  |
| PhotoAmt      | 23   | 14993 | 3.89     | 3.49    | 3      | 3.27     | 2.97    | 0     | 30    | 30    | 2.86  |
| AdoptionSpeed | 24   | 14993 | 2.52     | 1.18    | 2      | 2.55     | 1.48    | 0     | 4     | 4     | -0.16 |

The data was made up with 24 variables including Numerical and categorical data which is classified as tabular data as well as variables that include sentiment data and metadata that came

stored as .json files. There was also 14,993 observations in total meaning we are working with large data. The data was cleaned and data transformations were made to ensure interpretability in the models. The astrix shows these variables are non-numeric by default which means they are either character strings or ID's which means the describe function does not accurately describe these variables and cannot be properly interpreted yet.

Gender was initially stored as numeric data and was transformed to a factor variable which tells *glm()* to treat it as a categorical variable and automatically create dummy variables using one-hot encoding.

Adoption speed is our target variable which was initially coded to include five groups that categorised whether a pet was adopted within a day of their arrival, between one and seven days, eight and thirty days, thirty-one and ninety days and finally any pets that have been at the shelter for atleast 100 days (no pets were adopted between 90 and 99 days). Below is a histogram depicting the frequency of Adoption Speed for each category.

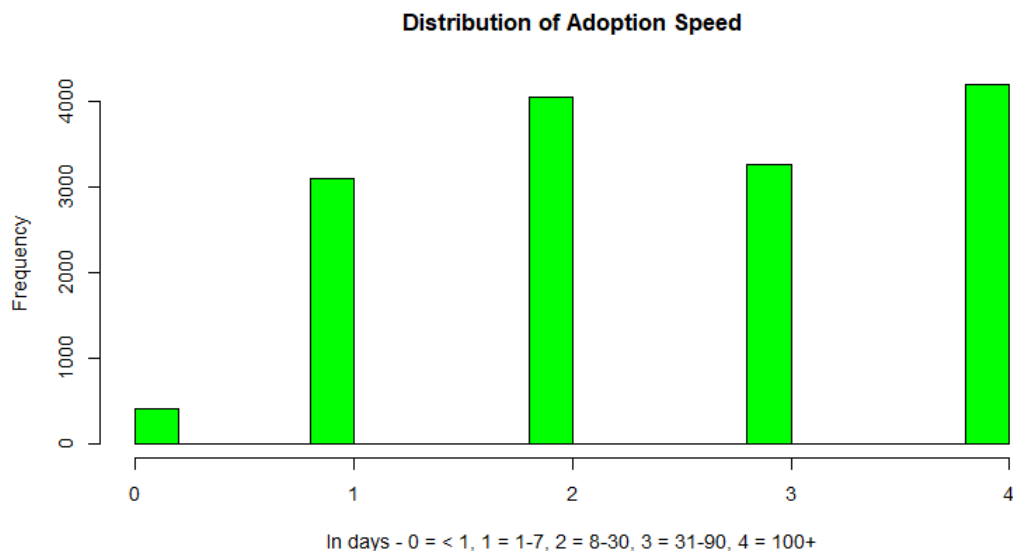


Figure 3.1: Distribution of the AdoptionSpeed variable before binarization

This was re-coded to a binary variable where 0 = thirty days or less and 1 = more than 30 days, shown below.

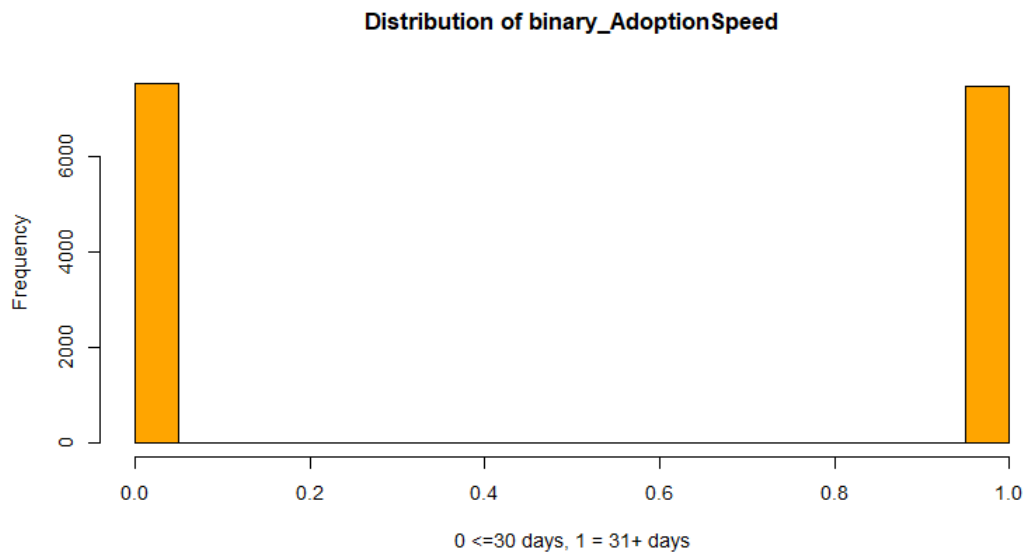


Figure 3.2: Distribution of the *binary\_AdoptionSpeed* variable

This shows an even distribution of categories to indicate whether a pet was adopted quickly or slowly using 30 days as the splitting point making it more balanced for fitting predictive models to draw conclusions from.

Investigating the Age variable, it is evident there is a high skew of the pets in the shelter. Skewed variables can have an outsized influence on logistic regression models. Transforming skewed variables help with better model fitting, more stable coefficient estimates and improved predictive performance.

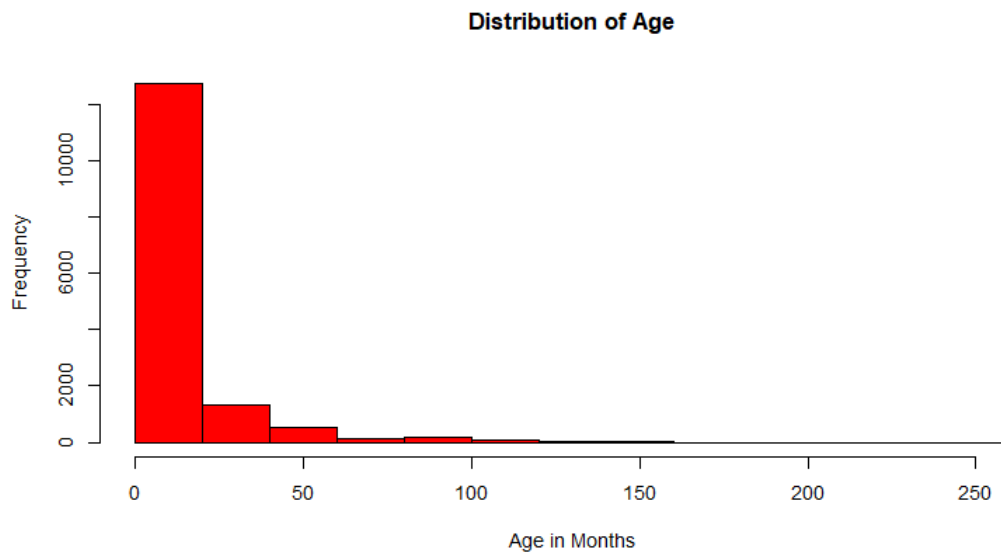
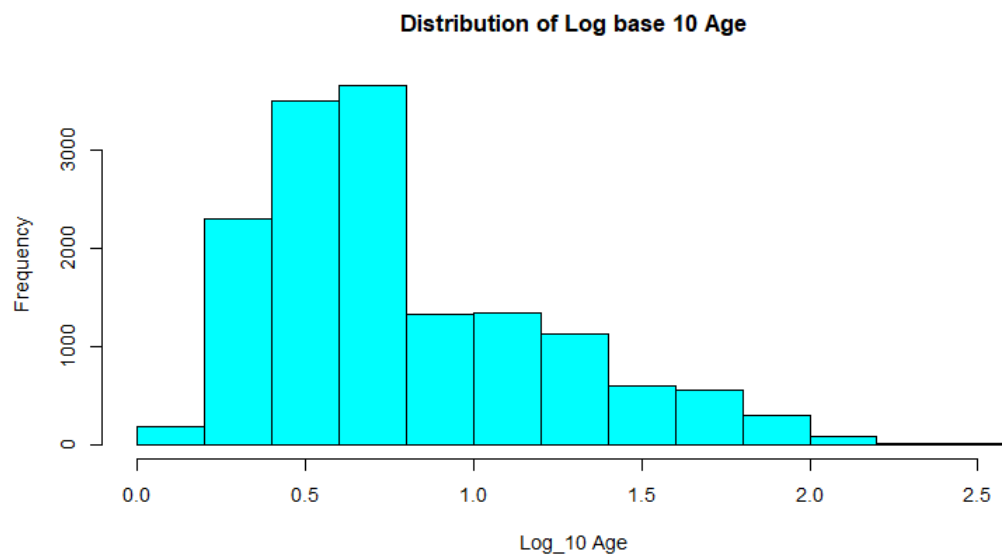


Figure 3.3: Distribution of the binary *AdoptionSpeed* variable

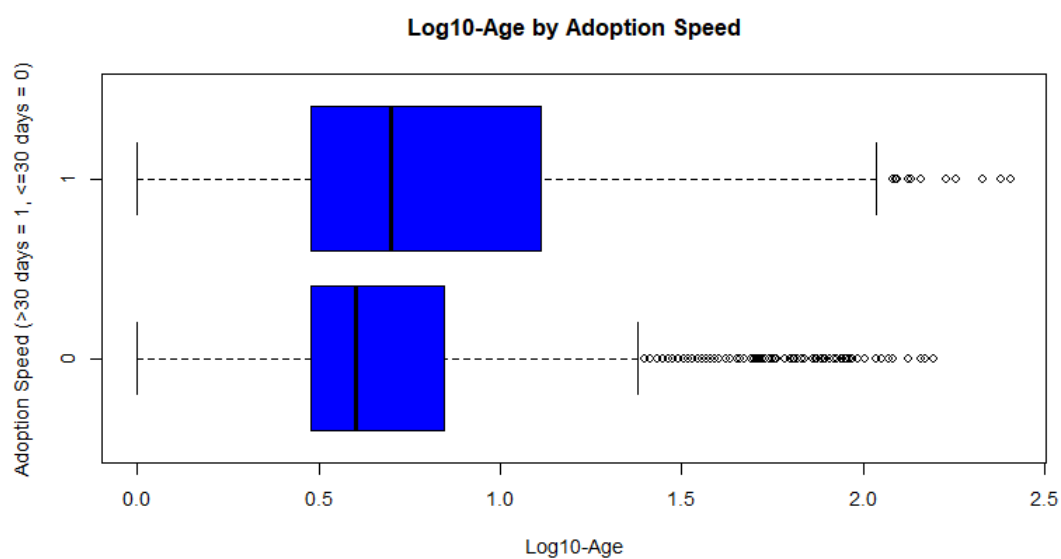
Primarily pets tend to be younger from this dataset and this was transformed using log base ten to help predictive power of the models. We can see the vast majority of pets in this shelter are younger which will give us something to think about when we consider some variables depicting whether they have been vaccinated and dewormed. It is likely a lot of them may not have received treatment given their youth since the largest portion of pets fill the column of 20 months or younger.

The transformed distribution of the age variable is shown here.

Figure 3.4: Distribution of  $\text{Log}_{10} \text{Age}$ 

This figure shows our new variable has a more normal distribution which can help the accuracy and interpretability of regression models being built.

Using the transformed variables we plot age and adoption speed against each other to get an idea of how they interact with each other.

Figure 3.5: Distribution of  $\text{Log}_{10} \text{Age}$



Clearly there is a relationship between them showing how younger pets are adopted quicker than older ones which makes sense as many people may opt for the option to raise their pets from young and give themselves the opportunity to train them properly from a young age. The median `log10_age` for pets adopted quickly (group 0) is lower than that of the slower adoption group. The spread and presence of higher-age outliers are more prevalent in the `>30` days group.

Next up the "Vaccinated" variable is transformed to a binary variable where 1 = they are definitely vaccinated and 0 = they are not or are unsure. The same transformation was made for "Dewormed", "Sterilized" as well as "Fee" showing 1 for having a fee to adopt the pet or 0 for no fee. Some further grouping was done with "Breed2" and "Color3" to make a compute simpler variables for understanding the models. If Breed2 has a secondary breed then they are given a "1" in a new "CrossBreed" variable, likewise "Tri-Colour" returns a "1" if the pet has a third colour and 0 for no third colour. The remaining Breed and Color variables are initially coded as numeric variables which are then mapped to their labels in the related .csv files given in the package. This created dummy variables that are then used in the modelling process to check for significance. There were a lot of different breeds to consider so further grouping was performed to exclude any breeds that appeared less than 100 times in the dataset. This was to focus more on breeds that had plenty of data to determine whether they were either more or less desirable for adoption. Any breeds that appeared less than 100 times were filtered into a "rare breed" option that also gets included in the modelling process. Any pet that only had a single colour was re-coded as "Solid" which is a term given to both cats and dogs that were not multi-coloured.

A new dataset was created to exclude some of the variables that were re-written as well as excluding the multimodal variables that were later added back in using the sentiment and metadata that was already created using Google Vision API and Google Natural Language API.

While fitting the initial regression model, `PetId` was specifically excluded as this variable encounters multicollinearity issues given it's direct relationship with the particular result showing whether the pet was adopted quickly or not. `PetID` was not excluded in this new dataset as it was required to map the multimodal data back into the dataset.

The step function in `r` was used to create a backward regression model that tests the significance of each variable to the model with *binaryAdoptionSpeed* as the target variable. While running this, the beginning AIC score (which does...) started at 19439.33 in the new dataset and went down to 19433.62 which is not a huge difference by any means. This model included the *type* of pet (`dog` = 1, `cat` = 2), *Gender* (1 = Male, 2 = Female, 3 = Mixed, if profile represents group of pets), *Furlength* (1 = Short, 2 = Medium, 3 = Long, 0 = Not Specified), *Health* (1 = Healthy,

2 = Minor Injury, 3 = Serious Injury, 0 = Not Specified), *Quantity* (Number of pets represented in profile), *Log10Age*, *VaccinatedBinary*, *DewormedBinary*, *SterilizedBinary*, *CrossBreed* (1 = crossbreed, 0 = purebreed), *Tri\_Colour* (1 = three coloured, 0 = 2 or less), *Fee\_binary*, *BreedName* (rare breed, mixed breed and any breed with one hundred or more observations), *ColorName.x* (Colour of the pet).

## Chapter 4

# Modelling methods

### 4.1 Logistic Regression

Logistic regression models were used in the project after deciding to transform our target variable to a binary variable. A backward regression approach also known as backward elimination was used to build a regression model as it starts with all the explanatory variables and iteratively removes the least significant ones until only meaningful predictors remain. This reduces the complexity of the model making the final model easier to interpret and use. This gives clearer insights into the relationship between significant predictors and the response variable (binary\_AdoptionSpeed in this case). It improves predictive accuracy by reducing non-informative or redundant variables which helps the risk of overfitting. The model uses statistical values such as p-values, adjusted  $R^2$  and AIC at each step, providing a clear, reproducible, and statistically justified methodology.

The logistic regression model estimates the probability  $P$  that an observation belongs to the positive class ( $Y = 1$ ), given predictors  $X_1, X_2, \dots, X_p$ :

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}}$$

where  $\beta_0$  is the intercept and  $\beta_j$  are the coefficients associated with each predictor  $X_j$ . The model outputs probabilities between 0 and 1, which are converted into binary predictions using a threshold (typically 0.5).

Logistic regression models the probability that a binary outcome  $Y$  equals 1, given a set of predictor variables  $X_1, X_2, \dots, X_p$ . The model takes the form:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-z}}, \quad \text{where } z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

In this project, the outcome  $Y$  represents whether a pet was adopted slowly (more than 30 days), and each  $X_j$  corresponds to a feature such as `Log10_Age`, `FurLength`, `VaccinatedBinary`, etc. The  $\beta_j$  values are the coefficients estimated during model training.

### Example:

Given the following estimated model:

$$z = -2.14 + 1.11 \cdot \text{Log10\_Age} + 0.52 \cdot \text{Type.x} - 0.22 \cdot \text{FurLength}$$

For a dog with:

- `Log10_Age` = 1.3
- `Type.x` = 1 (dog)
- `FurLength` = 1 (short fur)

We compute:

$$z = -2.14 + 1.11(1.3) + 0.52(1) - 0.22(1) = -0.397$$

$$P(Y = 1) = \frac{1}{1 + e^{0.397}} \approx 0.40$$

Since this probability is below 0.5, the model would predict that the pet is likely to be adopted **quickly** (classified as 0).

This illustrates how each variable influences the prediction, and how the logistic regression model converts these predictors into an interpretable probability.

## 4.2 Stepwise Selection Method

The step function in R was used to create a backward regression model that tests the significance of each variable while iteratively adding or removing predictor variables based on specific statistical criteria. It combines aspects of both forward selection and backward elimination. It adds relevant predictors and removes non-significant ones. This often results in simpler models that generalize well to new data by reducing redundancy and limiting the inclusion of non-informative predictors hence decreasing the risk of overfitting. This method is particularly useful at exploratory stages where the relationships between the variables are uncertain. This provides quick insights about important predictors. It is particularly useful when there is a large set of potential predictors and you want to quickly identify a smaller subset of the variables.

The Akaike Information Criterion (AIC) is calculated as:

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

where  $k$  is the number of model parameters and  $\hat{L}$  is the maximized likelihood of the model. Lower AIC values indicate better model quality, balancing goodness of fit with model complexity. In backward stepwise regression, AIC is used to iteratively remove variables that don't contribute significantly to the model's explanatory power.

## 4.3 LASSO Modelling

LASSO stands for Least Absolute Shrinkage and Selection Operator and is used as it's effective in feature selection and regularisation. Regularisation is used to shrink some coefficients to zero, effectively removing them from the model. This does not happen in the stepwise regression approach where coefficients are rarely zero. It automatically selects the most significant variables making a simpler, more interpretable model. This model helps prevent overfitting and improves the generalisation to new datasets.

$$\text{Minimize} \quad \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

- $y_i$  is the actual observed value of the response variable for observation  $i$ . This represents real-world data points being modelled.
- $\hat{y}_i^2$  is the predicted value of the response variable for observation  $i$ . This is the model's estimate of the response based on predictor variables.

- $n$  is the number of observations in the dataset. This defines the total dataset size used to fit the model.
- $\beta_j$  is the regression coefficient for the  $j$ th predictor variable. This represents the estimated influence of predictor  $j$  on the response variable.
- $p$  is the total number of predictor variables available in the dataset. This indicates how many predictors are initially considered for inclusion in the model.
- $\lambda$  is the penalty parameter (or shrinkage parameter) that controls the amount of regularization applied. Larger values of  $\lambda$  increase shrinkage and encourage simpler models by forcing more coefficients towards zero. When  $\lambda$  is zero, the model is just plain logistic regression. The best  $\lambda$  value balances model complexity and model performance.

$\sum_{i=1}^n (y_i - \hat{y}_i)^2$  This represents the sum of squared residuals (errors). It measures how well the model's predictions match the actual observed data. Minimizing this ensures a close fit to the data.  $\lambda \sum_{j=1}^p |\beta_j|$  This is called the L1 regularization penalty. It penalizes the absolute magnitude of regression coefficients to encourage sparsity (many coefficients become exactly zero). This helps in feature selection, effectively removing less important predictors from the model. Cross validation can be used to optimally choose the value of  $\lambda$ . This was done using the `cv.glmnet` package in R. K-fold cross-validation splits the training data into  $k$  folds where  $k$  is the number of subsets. Each fold is used as a validation set and the remaining  $k-1$  folds are used as the training set. The model is then fit at many different values of  $\lambda$ . Then the average prediction error (mean-squared error/deviance) is calculated for each  $\lambda$ . The best value for  $\lambda$  is then chosen to minimize the average error across all folds.

#### 4.3.1 Train-Test Splitting and Model Evaluation

To ensure robust and unbiased evaluation of model performance, the dataset was repeatedly broken into training and test sets using an 80/20 split. In each iteration, 80% of the observations were randomly sampled to form the training set and the remaining 20% makes up the test set. This process was repeated more than 100 times to assess the variability and stability of model performance across different data splits.

For each iteration a backward stepwise logistic regression model was refitted on the training subset. Predictions were made on the test subset using the trained model. Predicted probabilities were converted to binary class labels using a 0.5 threshold. This means that if the model predicted a probability greater than 0.5 for a pet being adopted slowly (more than 30 days), it was classified as that. The 0.5 threshold represents the standard decision boundary for binary classification when the cost of false positives and false negatives is considered equal. Performance

metrics were computed, specifically **Accuracy** which is the proportion of correctly predicted observations and **AUC** (Area Under the ROC Curve) which is a measure of the model's ability to discriminate between the two classes.

This repeated evaluation approach helps to mitigate the potential effects of random sampling and provides a more reliable estimate of the model's generalization performance. Both the backward stepwise and LASSO logistic regression models were evaluated using this same strategy, allowing for a fair and consistent comparison between their respective predictive capabilities.

## 4.4 Google Vision Api

For the metadata used in the modelling process, this was done using the Google Vision API which included `label_max` as our predictor variable. This uses pre-trained machine learning models to analyse images and understand their content. These models were trained on a vast amount of datasets and images that are designed to recognize and understand various visual elements, including objects, faces, landmarks, and text. `label_max_score` quantifies how sure the model is that the object detected in the images given matches the label it's defined. A higher score (1) indicates more confidence while a lower score (0) indicates the model is less certain.

## 4.5 Google Natural Language API

The Google Cloud Natural Language API uses machine learning to analyse text which provides insights on the sentiment data used in the modelling process. This API can extract information on unstructured data and can even detect the language used in the text. During sentiment analysis this API determines the overall sentiment of the text to show whether it's content is positive, negative or neutral as well as it's magnitude. The results of the analysis are returned in a structured format which can be integrated into the modelling process. Score and Magnitude were returned as new predictor variables. Score measures the overall sentiment of the data, indicating whether the text is positive, negative or neutral and returns a score between -1 and 1. Magnitude measure the strength or intensity of the sentiment measuring whether it's positive or negative. This gives values ranging from zero to infinity. Higher magnitude indicate stronger sentiment. These two values work together to provide a strong understanding of the sentiment given in the text.

### 4.5.1 Decision-Tree Models (CART)

These are flowchart-like diagrams that break down a problem into smaller parts, helping to visualize the potential outcomes of different choices.

Classification and Regression Trees (CART) divide the data step-by-step by choosing one feature

at a time. Each split is picked to group similar class labels together as much as possible in the resulting branches.

**Splitting criterion.** For a candidate split  $s$  that divides parent node  $t$  into children  $t_L$  and  $t_R$ , CART minimises the weighted impurity

$$\Delta I(s, t) = I(t) - \frac{|t_L|}{|t|} I(t_L) - \frac{|t_R|}{|t|} I(t_R), \quad (4.1)$$

where  $|t|$  is the number of samples in node  $t$  and  $I(\cdot)$  is a node-impurity measure. We use the *Gini index*

$$I_{\text{Gini}}(t) = \sum_{k=1}^K p_k(1 - p_k), \quad (4.2)$$

with  $p_k$  the proportion of class  $k$  in node  $t$ .

**Cost-complexity pruning.** To avoid over-fitting, CART grows a maximal tree and then prunes it by minimising the cost-complexity functional

$$R_\alpha(T) = R(T) + \alpha |T|, \quad (4.3)$$

where  $R(T)$  is the resubstitution error,  $|T|$  the number of terminal nodes, and  $\alpha \geq 0$  the complexity parameter (*cp*). Larger  $\alpha$  penalises tree size more heavily, yielding shallower, more generalisable trees.

**Hyper-parameter tuning.** We tuned  $\alpha$  over the grid  $\{0.0005, 0.0025, \dots, 0.02\}$  using repeated 5-fold cross-validation (3 repeats) and selected the value that maximised the mean area under the ROC curve (AUC). The optimal  $\hat{\alpha} = 0.0025$  (Figure 5.4).

**Variable importance.** CART provides a natural measure of variable importance: the total decrease in impurity contributed by each predictor, aggregated over all splits in the tree. Figure 5.6 shows the ten most influential features; *Mixed-breed status*, *Sterilisation* and *Age* dominate.

**Implementation.** All trees were trained in R 4.4 using the **rpart** package; cross-validation and tuning were handled by **caret**. The final pruned tree (Figure 5.5) contains  $|T| = 11$  terminal nodes and achieves AUC = 0.664 on the held-out test set (Figure 5.6).



## Chapter 5

# Results

The logistic regression results are shown here where **Log10\_Age** was one of the most statistically significant predictors ( $z = 20.80$ ,  $p < 2 \times 10^{-16}$ ). A positive coefficient ( $\beta = 1.11$ ) indicates that older pets are significantly more likely to experience slower adoption. Older pets (higher log10-age) are significantly more likely to be adopted slower. This is supported from the insight seen in the boxplot previously included.

Table 5.1: Logistic Regression Model Predicting Slow Adoption Speed (1 = adoption &gt; 30 days)

| Predictor                   | Estimate ( $\beta$ ) | Std. Error | z-value | p-value    |
|-----------------------------|----------------------|------------|---------|------------|
| Intercept                   | -2.142               | 0.312      | -6.871  | < 0.001*** |
| Type.x (Dog)                | 0.518                | 0.099      | 5.248   | < 0.001*** |
| Gender (Male)               | 0.248                | 0.038      | 6.518   | < 0.001*** |
| Gender (Unknown)            | 0.339                | 0.071      | 4.789   | < 0.001*** |
| FurLength                   | -0.223               | 0.035      | -6.419  | < 0.001*** |
| Health                      | 0.334                | 0.089      | 3.752   | < 0.001*** |
| Quantity                    | 0.086                | 0.017      | 5.089   | < 0.001*** |
| Log10_Age                   | 1.109                | 0.053      | 20.801  | < 0.001*** |
| Vaccinated (Yes)            | 0.185                | 0.050      | 3.728   | < 0.001*** |
| Dewormed (Yes)              | -0.121               | 0.045      | -2.664  | 0.008**    |
| Sterilized (Yes)            | 0.356                | 0.052      | 6.849   | < 0.001*** |
| CrossBreed                  | 0.129                | 0.041      | 3.165   | 0.002**    |
| Tri_Colour                  | -0.076               | 0.043      | -1.782  | 0.075      |
| Fee (Yes)                   | 0.193                | 0.049      | 3.924   | < 0.001*** |
| Breed: Domestic Long Hair   | -0.554               | 0.245      | -2.265  | 0.023*     |
| Breed: Domestic Medium Hair | -0.195               | 0.215      | -0.906  | 0.365      |
| Breed: Domestic Short Hair  | -0.197               | 0.210      | -0.941  | 0.347      |
| Breed: Golden Retriever     | -0.726               | 0.292      | -2.482  | 0.013*     |
| Breed: Labrador Retriever   | -0.028               | 0.272      | -0.103  | 0.918      |
| Breed: Mixed Breed          | 0.773                | 0.231      | 3.347   | < 0.001*** |
| Breed: Persian              | -0.937               | 0.258      | -3.633  | < 0.001*** |
| Breed: Poodle               | -0.980               | 0.287      | -3.411  | < 0.001*** |
| Breed: Rare Breed           | -0.339               | 0.220      | -1.537  | 0.124      |
| Breed: Shih Tzu             | -1.095               | 0.280      | -3.913  | < 0.001*** |
| Breed: Siamese              | -0.889               | 0.248      | -3.593  | < 0.001*** |
| Breed: Tabby                | -0.319               | 0.235      | -1.356  | 0.175      |
| Breed: Terrier              | 0.361                | 0.281      | 1.283   | 0.200      |
| Color: Brown                | 0.005                | 0.044      | 0.124   | 0.901      |
| Color: Cream                | -0.309               | 0.078      | -3.991  | < 0.001*** |
| Color: Golden               | -0.168               | 0.076      | -2.202  | 0.028*     |
| Color: Gray                 | -0.176               | 0.088      | -1.997  | 0.046*     |
| Color: White                | -0.303               | 0.088      | -3.434  | < 0.001*** |
| Color: Yellow               | 0.166                | 0.088      | 1.879   | 0.060      |

Significance codes: \*\*\* $p < 0.001$ , \*\* $p < 0.01$ , \* $p < 0.05$ ,  $p < 0.1$

Null deviance: 20777 (df = 14987), Residual deviance: 19368 (df = 14955), AIC: 19434

This model shows several covariates to be significant with a p value of 0.05 or less below Type.x: Dogs were more likely to be adopted slowly ( $\beta = 0.52$ ,  $p < 0.001$ ). Gender: Male pets (Gender2) and unknown gender (Gender3) had higher odds of slower adoption compared to females. This indicates more of a demand for female pets to adopters. FurLength: Longer fur was associated

with faster adoptions ( $\beta = -0.22$ ). **Health:** Poorer health was correlated with slower adoption ( $\beta = 0.33$ ). **Quantity:** Pets listed in larger quantities were marginally more likely to take longer to be adopted. **SterilizedBinary:** Pets that were sterilized were more likely to be adopted quickly ( $\beta = 0.36$ ,  $p < 0.001$ ). **DewormedBinary** ( $\beta = -0.12$ ,  $p = 0.0077$ ). Pets that have been dewormed tend to be adopted faster. **Breed and Color:** Specific breeds such as *Shih Tzu*, *Persian*, and *Poodle* had significantly lower adoption delay odds, while mixed breeds had higher odds ( $\beta = 0.77$ ,  $p < 0.001$ ). Color-wise, cream, golden, gray, and white all had significantly negative coefficients, suggesting these colors were adopted faster.

Positive coefficients mean this variable gives a stronger likelihood of being adopted slower while negative ones show faster adoption. From this model we can tell that based on our data; younger pets are adopted quicker, the same goes for female pets, Longer fur, healthy pets, individually listed pets, sterilised pets, dewormed, Shit-Zhu's, Poodle's, Persian's, Siamese, Golden Retriever's, Domestic Long Hair, Cream coloured, White, Gray and Golden coloured pets. Overall, the model achieved a residual deviance of 19,368 on 14,955 degrees of freedom, with an AIC of 19,434, indicating a good fit given the size of the dataset and number of predictors. These insights could directly inform shelter practices such as strategic marketing and focused adoption campaigns could aim to enhance adoption rates and reduce shelter stays.

Some of the dummy variables for breed and colour were insignificant here but were kept in the model as further grouping became more difficult. These were kept in the model as the multimodal data has not been included yet and could make some variables that are close to significance to be significant. This model remains to be quite large so a penalized regression approach was done to see try simplify it further. The LASSO model is given below;

Table 5.2: LASSO Logistic Regression Model Predicting Slow Adoption Speed

| Predictor                   | Coefficient ( $\beta$ ) |
|-----------------------------|-------------------------|
| Intercept                   | -16.276                 |
| Type.x (Dog)                | 0.441                   |
| Gender1 (Female)            | -0.241                  |
| Gender3 (Unknown)           | 0.091                   |
| MaturitySize                | -0.036                  |
| FurLength                   | -0.215                  |
| Health                      | 0.316                   |
| Quantity                    | 0.084                   |
| State                       | 0.0003                  |
| PhotoAmt                    | 0.0001                  |
| Log10_Age                   | 1.090                   |
| Vaccinated (Yes)            | 0.171                   |
| Dewormed (Yes)              | -0.106                  |
| Sterilized (Yes)            | 0.357                   |
| CrossBreed                  | 0.125                   |
| Tri_Colour                  | -0.014                  |
| Fee (Yes)                   | 0.185                   |
| Breed: Domestic Long Hair   | -0.335                  |
| Breed: Domestic Medium Hair | 0 (removed)             |
| Breed: Domestic Short Hair  | 0 (removed)             |
| Breed: Golden Retriever     | -0.554                  |
| Breed: Labrador Retriever   | 0.100                   |
| Breed: Mixed Breed          | 0.907                   |
| Breed: Persian              | -0.705                  |
| Breed: Poodle               | -0.842                  |
| Breed: Rare Breed           | -0.162                  |
| Breed: Shih Tzu             | -0.930                  |
| Breed: Siamese              | -0.628                  |
| Breed: Tabby                | -0.100                  |
| Breed: Terrier              | 0.493                   |
| Color: Brown                | 0 (removed)             |
| Color: Cream                | -0.319                  |
| Color: Golden               | -0.164                  |
| Color: Gray                 | -0.182                  |
| Color: White                | -0.341                  |
| Color: Yellow               | 0.169                   |
| Secondary Color: Cream      | -0.093                  |
| Secondary Color: Golden     | 0.027                   |
| Secondary Color: Gray       | -0.029                  |
| Secondary Color: Solid      | 0.098                   |
| Secondary Color: White      | 0.044                   |
| Secondary Color: Yellow     | 0.093                   |

Note: Coefficients set to zero were removed by LASSO regularization.

Shown in this figure we can see that Breed: Domestic Medium/Short Hair and Color: Brown were removed. Overall the model is very similar to the stepwise approach with a slight improvement. These two models were then compared against eachother using a test sample from the dataset as well as comparing the confusion matrices for each model.

To perform a LASSO model, the best performing lambda is found through trial and is found here;

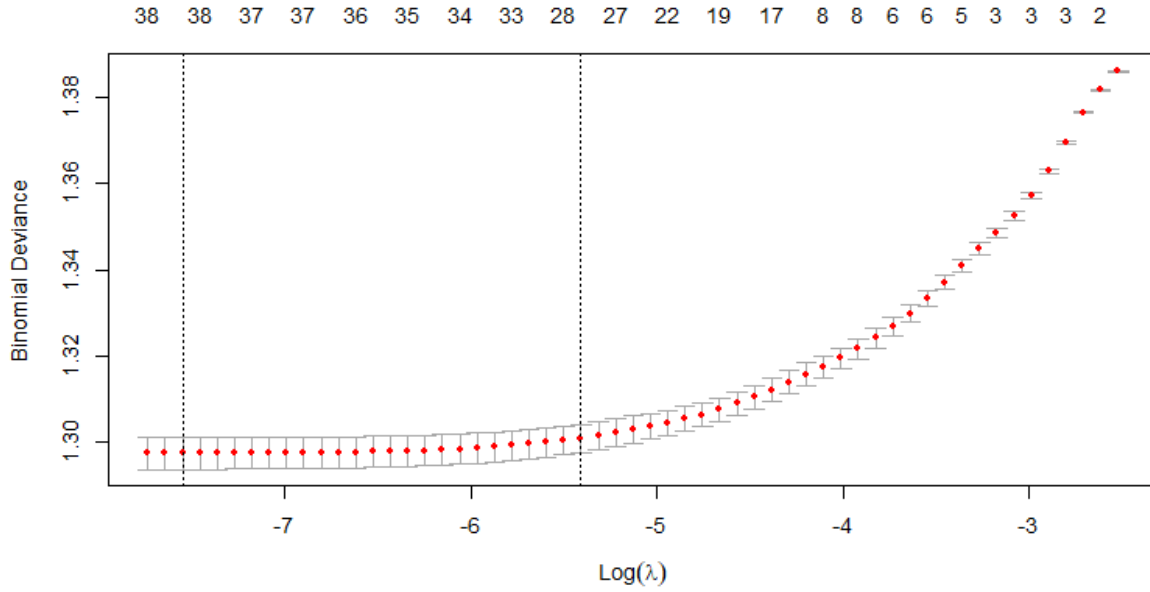


Figure 5.1: Non-zero coefficients from the LASSO logistic-regression model. Positive values indicate factors associated with slower adoption ( $> 30$  days); negative values the opposite. Variables shrunk to zero (e.g., *Breed: Domestic Medium Hair*, *Color: Brown*) are omitted.

Below is the confusion matrix for the stepwise approach tested on a subset of the dataset;

Table 5.3: Confusion Matrix for Stepwise Logistic Regression Model

| Predicted / Actual        | 0 (Adopted $< 31$ days) | 1 (Adopted $> 30$ days) |
|---------------------------|-------------------------|-------------------------|
| 0 (Predicted $< 31$ days) | 981                     | 621                     |
| 1 (Predicted $> 30$ days) | 522                     | 875                     |

From this we can see that 981 were correctly predicted as true negatives meaning they were correctly predicted to be adopted quickly (within 30 days). 875 were correctly predicted as true positives meaning they accurately predicted 875 slow adoptions (more than 30 days). Conversely, 621 false positives and 522 false negatives were predicted. The accuracy of this model is given

here;

Table 5.4: Accuracy of Backward Stepwise Logistic Regression Model

| Model                                 | Accuracy |
|---------------------------------------|----------|
| Backward Stepwise Logistic Regression | 61.89%   |

This backward stepwise logistic regression model correctly classified 61.89% of test cases.

To compare, the LASSO confusion matrix is seen here, followed by the accuracy of the LASSO logistic regression model;

Table 5.5: Confusion Matrix for LASSO Logistic Regression Model

| Predicted / Actual      | 0 (Adopted < 31 days) | 1 (Adopted > 30 days) |
|-------------------------|-----------------------|-----------------------|
| 0 (Predicted < 31 days) | 976                   | 613                   |
| 1 (Predicted > 30 days) | 527                   | 883                   |

Table 5.6: Accuracy of LASSO Logistic Regression Model

| Model                     | Accuracy |
|---------------------------|----------|
| LASSO Logistic Regression | 61.99%   |

Since there is such a subtle difference between the two models, this test was performed more than 100 times and the average scores were saved to compare the overall difference. The average accuracy and AUC results from this test are given here;

Table 5.7: Average Accuracy Comparison between LASSO and Stepwise Models (over 100 iterations)

| Model                                 | Average Accuracy |
|---------------------------------------|------------------|
| LASSO Logistic Regression             | 62.77%           |
| Backward Stepwise Logistic Regression | 62.40%           |

And here;

Table 5.8: Average AUC Comparison between LASSO and Stepwise Models (over 100 iterations)

| Model                                 | Average AUC |
|---------------------------------------|-------------|
| LASSO Logistic Regression             | 0.674       |
| Backward Stepwise Logistic Regression | 0.669       |

These tables show similar results where the LASSO Logistic Regression Model (62.77%) is slightly more accurate than the backward Stepwise Logistic Regression Model (62.4%).

The ROC curve depicting the stepwise regression model and the LASSO regression model is seen here;

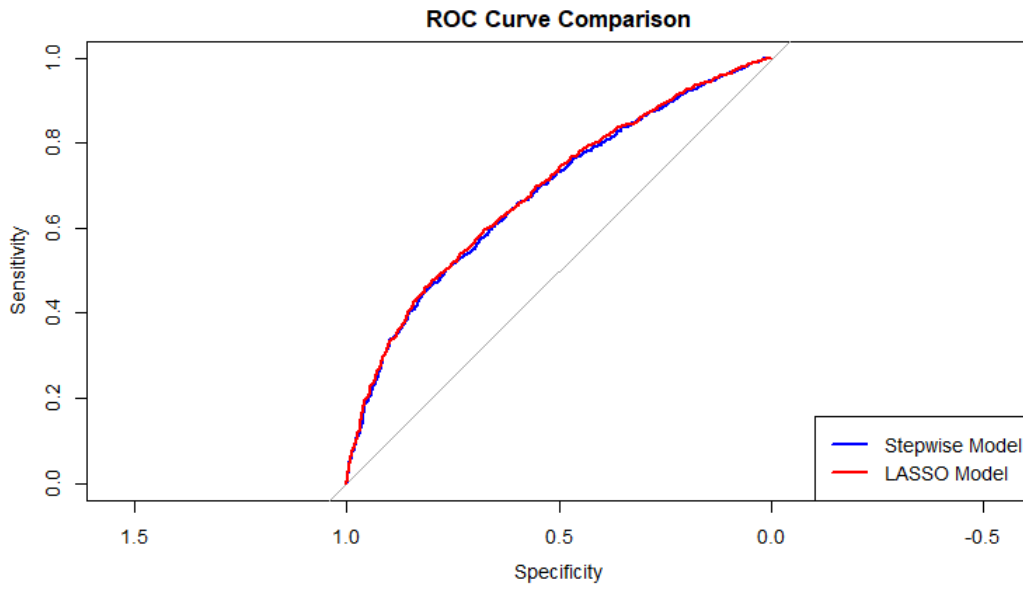


Figure 5.2: ROC Curve Comparing Stepwise and LASSO Model

The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR), where:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

The AUC (Area Under the Curve) measures the entire two-dimensional area underneath the ROC curve, with values closer to 1.0 indicating better discriminative ability.

After adding in the multimodal data which contained three new variables - Score, Magnitude and Label\_max\_score. A new backward logistic regression model was created to include these which increased the accuracy of the prediction model. Firstly, a stepwise selection model was built which gave

Table 5.9: Stepwise Logistic Regression Model Using Multimodal Features

| Predictor                     | Estimate ( $\beta$ ) | Std. Error | z-value | p-value                | Signif. |
|-------------------------------|----------------------|------------|---------|------------------------|---------|
| (Intercept)                   | 5.7517               | 1.2076     | 4.763   | $1.91 \times 10^{-6}$  | ***     |
| Type.x                        | 0.6170               | 0.1069     | 5.772   | $7.83 \times 10^{-9}$  | ***     |
| Gender2                       | 0.2817               | 0.0394     | 7.157   | $8.25 \times 10^{-13}$ | ***     |
| Gender3                       | 0.3392               | 0.0742     | 4.573   | $4.81 \times 10^{-6}$  | ***     |
| FurLength                     | -0.2221              | 0.0362     | -6.139  | $8.32 \times 10^{-10}$ | ***     |
| Health                        | 0.3619               | 0.0937     | 3.863   | $1.12 \times 10^{-4}$  | ***     |
| Quantity                      | 0.0817               | 0.0179     | 4.569   | $4.89 \times 10^{-6}$  | ***     |
| PhotoAmt                      | 0.0216               | 0.0055     | 3.919   | $8.89 \times 10^{-5}$  | ***     |
| Log10_Age                     | 1.1251               | 0.0559     | 20.113  | $< 2 \times 10^{-16}$  | ***     |
| VaccinatedBinary              | 0.1642               | 0.0512     | 3.210   | $1.33 \times 10^{-3}$  | **      |
| DewormedBinary                | -0.0738              | 0.0476     | -1.549  | 0.1215                 | .       |
| SterilizedBinary              | 0.3781               | 0.0538     | 7.031   | $2.06 \times 10^{-12}$ | ***     |
| CrossBreed                    | 0.1402               | 0.0424     | 3.306   | $9.48 \times 10^{-4}$  | ***     |
| Tri.Colour                    | -0.1037              | 0.0447     | -2.323  | 0.0202                 | *       |
| Fee.binary                    | 0.2111               | 0.0509     | 4.148   | $3.36 \times 10^{-5}$  | ***     |
| BreedNameDomestic Long Hair   | -0.4871              | 0.2565     | -1.899  | 0.0576                 | .       |
| BreedNameDomestic Medium Hair | -0.2442              | 0.2243     | -1.089  | 0.2762                 | .       |
| BreedNameDomestic Short Hair  | -0.2972              | 0.2181     | -1.363  | 0.1729                 | .       |
| BreedNameGolden Retriever     | -0.9556              | 0.3066     | -3.117  | 0.0018                 | **      |
| BreedNameLabrador Retriever   | -0.1843              | 0.2816     | -0.654  | 0.5128                 | .       |
| BreedNameMixed Breed          | 0.6099               | 0.2401     | 2.540   | 0.0111                 | *       |
| BreedNamePersian              | -1.0640              | 0.2709     | -3.927  | $8.60 \times 10^{-5}$  | ***     |
| BreedNamePoodle               | -1.1986              | 0.2990     | -4.008  | $6.13 \times 10^{-5}$  | ***     |
| BreedNameRare Breed           | -0.4940              | 0.2296     | -2.152  | 0.0314                 | *       |
| BreedNameShih Tzu             | -1.3399              | 0.2919     | -4.591  | $4.42 \times 10^{-6}$  | ***     |
| BreedNameSiamese              | -1.0420              | 0.2587     | -4.027  | $5.64 \times 10^{-5}$  | ***     |
| BreedNameTabby                | -0.3783              | 0.2440     | -1.550  | 0.1211                 | .       |
| BreedNameTerrier              | 0.1786               | 0.2903     | 0.615   | 0.5384                 | .       |
| ColorName.xBrown              | -0.0090              | 0.0452     | -0.199  | 0.8422                 | .       |
| ColorName.xCream              | -0.3358              | 0.0802     | -4.187  | $2.82 \times 10^{-5}$  | ***     |
| ColorName.xGolden             | -0.1631              | 0.0795     | -2.051  | 0.0403                 | *       |
| ColorName.xGray               | -0.2528              | 0.0926     | -2.730  | 0.0063                 | **      |
| ColorName.xWhite              | -0.4318              | 0.0924     | -4.676  | $2.93 \times 10^{-6}$  | ***     |
| ColorName.xYellow             | 0.1207               | 0.0921     | 1.311   | 0.1899                 | .       |
| score                         | 0.2629               | 0.0655     | 4.016   | $5.91 \times 10^{-5}$  | ***     |
| magnitude                     | -0.0338              | 0.0093     | -3.633  | 0.0003                 | ***     |
| label_max_score               | -8.2542              | 1.2311     | -6.705  | $2.02 \times 10^{-11}$ | ***     |

Significance codes: \*\*\* $p < 0.001$ , \*\* $p < 0.01$ , \* $p < 0.05$ ,  $p < 0.1$ 

Null deviance: 19554 (df = 14106), Residual deviance: 18086 (df = 14070), AIC = 18160

Lasso model for multimodal



Table 5.10: LASSO Logistic Regression Model Coefficients (Multimodal Features)

| Predictor                   | Coefficient ( $\beta$ ) |
|-----------------------------|-------------------------|
| (Intercept)                 | 5.925                   |
| Type.x                      | 0.484                   |
| Gender2                     | 0.267                   |
| Gender3                     | 0.299                   |
| MaturitySize                | -0.028                  |
| FurLength                   | -0.186                  |
| Health                      | 0.316                   |
| Quantity                    | 0.096                   |
| PhotoAmt                    | 0.016                   |
| Log10_Age                   | 1.038                   |
| VaccinatedBinary            | 0.125                   |
| DewormedBinary              | -0.051                  |
| SterilizedBinary            | 0.427                   |
| CrossBreed                  | 0.118                   |
| Tri_Colour                  | -0.047                  |
| Fee_binary                  | 0.180                   |
| BreedNameDomestic Long Hair | -0.158                  |
| BreedNameGolden Retriever   | -0.682                  |
| BreedNameLabrador Retriever | -0.039                  |
| BreedNameMixed Breed        | 0.724                   |
| BreedNamePersian            | -0.755                  |
| BreedNamePoodle             | -1.088                  |
| BreedNameRare Breed         | -0.207                  |
| BreedNameShih Tzu           | -1.074                  |
| BreedNameSiamese            | -0.701                  |
| BreedNameTabby              | -0.136                  |
| BreedNameTerrier            | 0.274                   |
| ColorName.xCream            | -0.333                  |
| ColorName.xGolden           | -0.120                  |
| ColorName.xGray             | -0.210                  |
| ColorName.xWhite            | -0.396                  |
| ColorName.xYellow           | 0.165                   |
| ColorName.yCream            | -0.183                  |
| ColorName.yGolden           | 0.021                   |
| ColorName.yGray             | -0.106                  |
| ColorName.ySolid            | 0.062                   |
| ColorName.yWhite            | -0.067                  |
| ColorName.yYellow           | 0.039                   |
| score                       | 0.212                   |
| magnitude                   | -0.031                  |
| label_max_score             | -8.314                  |

After running 100 iterations comparison between the stepwise and lasso model and logging the

average scores, the LASSO model is slightly more accurate which makes sense as the LASSO model shrinks the insignificant variables to zero so the model only uses significant variables to predict adoption speed. The results from the 100 iterations is shown below;

Table 5.11: Average Accuracy and AUC over 100 Iterations for Stepwise and LASSO Models

| Model                        | Accuracy | AUC   |
|------------------------------|----------|-------|
| Stepwise Logistic Regression | 63.37%   | 0.678 |
| LASSO Logistic Regression    | 63.47%   | 0.678 |

this is the combined confusion matrix

Table 5.12: Comparison of Confusion Matrices: Stepwise vs LASSO Models

| Predicted / Actual     | Stepwise |     | LASSO |     |
|------------------------|----------|-----|-------|-----|
|                        | 0        | 1   | 0     | 1   |
| 0 (Predicted 30 days)  | 983      | 574 | 981   | 578 |
| 1 (Predicted >30 days) | 461      | 804 | 463   | 800 |

In this test we see the LASSO model actually marginally under-performs compared to the stepwise model but the difference is marginal so both models are good for predicting the adoption speed of pets.

This table again includes some insignificant dummy variables so a LASSO model was built to increase accuracy. The model shows that all three new variables were significant to the predictive model process which indicates a more accurate prediction model.

The area on both models again show how close the LASSO model is to the previous model and so is the stepwise model.

To understand whether the sparsity induced by LASSO actually improves discrimination, we compared the cross-validated AUC of the LASSO and a stepwise logistic model (Figure 5.3). Median AUC differs by only 0.001, confirming that model parsimony rather than raw performance gain motivated our choice of LASSO.

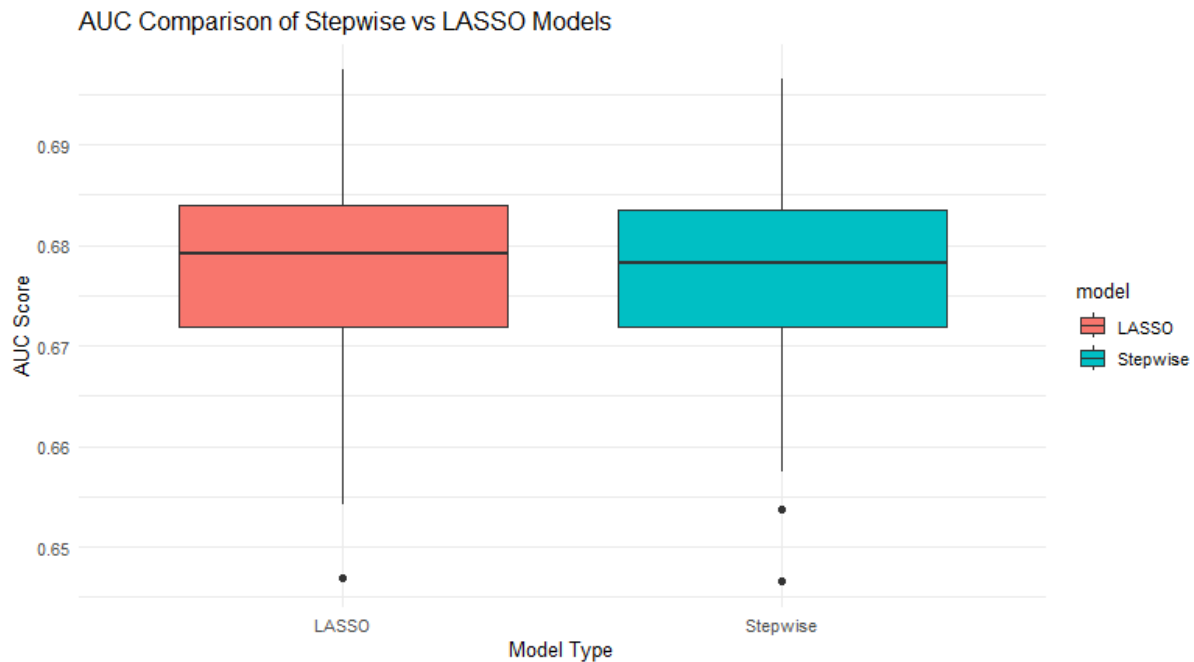


Figure 5.3: Distribution of cross-validated AUC for the LASSO and stepwise logistic-regression models (10×5-fold CV). Horizontal lines mark the median; whiskers show the full range.

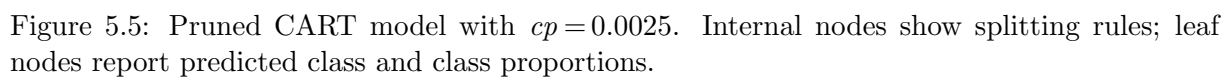
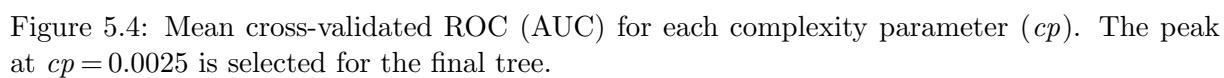
### Results of CART with Hyperparameter Tuning

Table 5.13: Performance of CART models

| Model<br>Bal. Acc. | Data split | $c_p$  | Accuracy | AUC   | Sens  | Spec  |
|--------------------|------------|--------|----------|-------|-------|-------|
| Pruned CART        | Test       | 0.010  | 0.603    | 0.605 | 0.811 | 0.389 |
| 0.600              |            |        |          |       |       |       |
| Tuned CART         | CV         | 0.0025 | –        | 0.663 | 0.673 | 0.584 |
| –                  |            |        |          |       |       |       |

We trained a CART model using the `rpart` package and tuned its complexity parameter ( $c_p$ ) via repeated 5-fold cross-validation in `caret`, scanning  $c_p$  values from 0.0005 to 0.02. The optimal  $c_p$  was 0.0025, yielding a cross-validated AUC of 0.6643 ( $\pm 0.0098$ ), sensitivity 0.67 and specificity 0.59 (Table 5.13; Figure 5.4).

The most important predictors were Mixed-Breed status, Sterilization, and Age, indicating that non-linear interactions among these features strongly influence adoption speed. This model improves AUC by 6 percentage points over the baseline  $c_p=0.01$  tree and provides interpretable splits for targeted interventions (e.g. prioritizing sterilization campaigns for young mixed-breed animals).



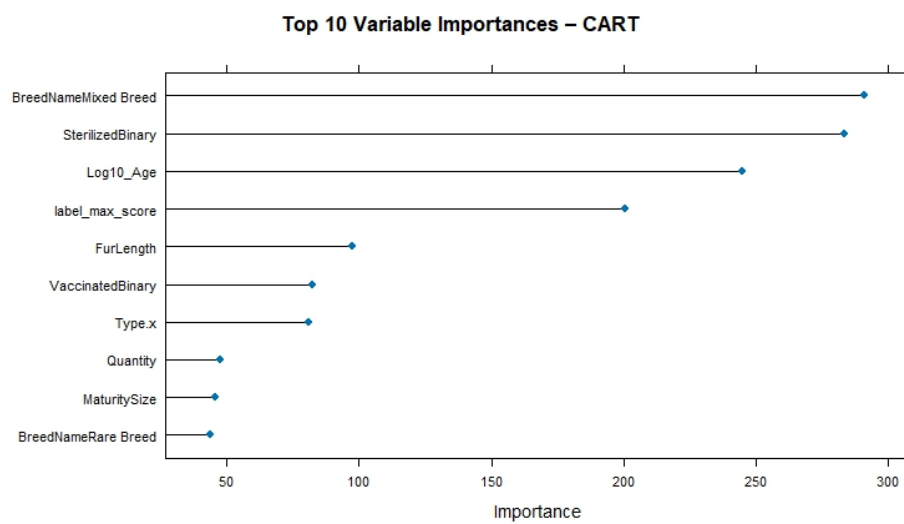


Figure 5.6: Top-10 variable importances from the final CART model, measured by total Gini-impurity reduction.

## Chapter 6

# Discussion, Conclusions and further work

### 6.1 Discussion

This study aimed to identify key predictors influencing pet adoption speed and assess the effectiveness of traditional logistic regression models in understanding adoption patterns in Malaysia. We developed robust logistic models by transforming and analyzing tabular data, uncovering meaningful relationships between adoption speed and several predictors, including *Log10Age*, *VaccinatedBinary*, *SterilizedBinary*, and *Health*. While logistic regression revealed key predictors, multimodal data (e.g., images and text) will be explored in future work to enhance prediction accuracy. Advanced models such as neural networks and hybrid approaches will also be considered. The findings highlighted that practical measures such as ensuring pets are vaccinated and sterilized significantly enhance their likelihood of quick adoption. Age also played a crucial factor, with older pets showing higher adoption probabilities. Interestingly, fee and fur length had subtle but statistically significant impacts, with higher fees and longer fur slightly reducing adoption rates.

The refined Model 2 offered better interpretability and fit compared to Model 1 by excluding meaningless predictors like *Breed1* and *Color1*. However, some issues remain. For example, the current analysis primarily relies on tabular data, which might not fully capture other adoption factors such as visual appearance (images) or descriptive characteristics (textual descriptions). Additionally, climate-specific influences on adoption preferences in Malaysia need further exploration.

## 6.2 Conclusion

This research successfully identified significant predictors of pet adoption speed in Malaysia using logistic regression models. The study found that pets that are vaccinated, sterilized, and healthy are significantly more likely to be adopted quickly. Age plays a prominent role, with older pets often adopted faster. Fee and fur length have smaller but noteworthy influences. These insights can guide shelter practices to improve adoption outcomes, such as highlighting the need for pet vaccination and sterilization status and tailoring adoption campaigns based on pet characteristics. While logistic regression provided valuable insights, the analysis underscored the need for more advanced methods to account for the multimodal nature of the dataset and interactions between variables.

## 6.3 Further Work

While this study focused on tabular data, future work will incorporate multimodal approaches, including image and text data, to improve predictive accuracy. Convolutional Neural Networks (CNNs) will analyze image features like pet appearance, while Natural Language Processing (NLP) methods will extract insights from textual descriptions. This shows nonlinear relationships and context-specific variables that are not captured by logistic regression alone. However, data preprocessing, computational cost, and ensuring model interpretability remain unaddressed. Incorporate image data using convolutional neural networks (CNNs) to analyze visual appeal factors. Analyze text descriptions with natural language processing (NLP) techniques to assess the impact of wording on adoption speed. Hybrid models combine logistic regression with neural networks to create hybrid models that balance interpretability and predictive power. An investigation into whether ensemble techniques, such as boosting could improve model accuracy and strength. Cultural and contextual factors explore how Malaysian cultural values and climate-specific factors influence adoption trends. Comparative studies with datasets from other regions can further contextualize the findings. Explainability in models as more complex ones are explored, implement tools like SHAP (Shapley Additive Explanations) to ensure transparency in predictions. By extending this work, the study can move beyond tabular data, providing a meaningful understanding of adoption trends and actionable strategies to improve pet adoption rates in Malaysia and beyond.

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